

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: (II)

Problem Set 1 - Solutions

(Sufficiency, Minimal Sufficiency, Unbiasedness)

[1] Finding Sufficient Statistics (Neyman-Fisher Factorization):

The Factorization Theorem states that $T(\underline{X})$ is sufficient for θ if the joint pdf can be written as $f_{\theta}(\underline{x}) = g_{\theta}(T(\underline{x}))h(\underline{x})$.

(a) $f_{\alpha}(x) = \frac{1}{\alpha} \exp\left(-\frac{x}{\alpha}\right) \mathbb{I}(x > 0)$.

The joint pdf is:

$$f_{\alpha}(\underline{x}) = \prod_{i=1}^n \frac{1}{\alpha} e^{-x_i/\alpha} = \frac{1}{\alpha^n} e^{-\frac{1}{\alpha} \sum_{i=1}^n x_i}$$

Let $h(\underline{x}) = 1$ and $g_{\alpha}(t) = \frac{1}{\alpha^n} e^{-t/\alpha}$. Thus, $T(\underline{X}) = \sum_{i=1}^n X_i$ is sufficient for α .

(b) $f_{\beta}(x) = \exp(-(x - \beta)) \mathbb{I}(x > \beta)$.

The joint pdf is:

$$f_{\beta}(\underline{x}) = \prod_{i=1}^n e^{-(x_i - \beta)} \mathbb{I}(x_i > \beta) = e^{-\sum x_i} e^{n\beta} \mathbb{I}(\min x_i > \beta)$$

Let $h(\underline{x}) = e^{-\sum x_i}$ and $g_{\beta}(t) = e^{n\beta} \mathbb{I}(t > \beta)$. Thus, $T(\underline{X}) = X_{(1)}$ (the minimum order statistic) is sufficient for β .

(c) $f_{\alpha, \beta}(x) = \frac{1}{\alpha} \exp\left(-\frac{x - \beta}{\alpha}\right) \mathbb{I}(x > \beta)$.

The joint pdf is:

$$\begin{aligned} f_{\alpha, \beta}(\underline{x}) &= \frac{1}{\alpha^n} \exp\left(-\frac{\sum x_i - n\beta}{\alpha}\right) \mathbb{I}(x_{(1)} > \beta) \\ &= \left[\frac{1}{\alpha^n} e^{-\frac{\sum x_i}{\alpha}} e^{\frac{n\beta}{\alpha}} \mathbb{I}(x_{(1)} > \beta) \right] \cdot 1 \end{aligned}$$

Factorization implies $T(\underline{X}) = (\sum_{i=1}^n X_i, X_{(1)})$ is jointly sufficient for (α, β) .

(d) $f_{\mu, \sigma}(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right)$.

The joint pdf is:

$$f_{\mu, \sigma}(\underline{x}) = \left(\prod \frac{1}{x_i} \right) \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left(-\frac{1}{2\sigma^2} \sum (\log x_i - \mu)^2\right)$$

Expanding the square: $\sum (\log x_i - \mu)^2 = \sum (\log x_i)^2 - 2\mu \sum \log x_i + n\mu^2$. Let $h(\underline{x}) = \prod \frac{1}{x_i}$. The rest depends on the data only through $\sum \log x_i$ and $\sum (\log x_i)^2$. Thus, $T(\underline{X}) = (\sum_{i=1}^n \log X_i, \sum_{i=1}^n (\log X_i)^2)$ is sufficient.

(e) $f_{\theta}(x) = \frac{1}{\theta} \mathbb{I}(-\theta/2 \leq x \leq \theta/2)$.

The joint pdf is:

$$f_{\theta}(\underline{x}) = \frac{1}{\theta^n} \prod \mathbb{I}\left(-\frac{\theta}{2} \leq x_i \leq \frac{\theta}{2}\right) = \frac{1}{\theta^n} \mathbb{I}\left(x_{(1)} \geq -\frac{\theta}{2} \text{ and } x_{(n)} \leq \frac{\theta}{2}\right)$$

This simplifies to checking that $\theta \geq 2x_{(n)}$ and $\theta \geq -2x_{(1)}$, which is equivalent to $\theta \geq 2 \max |x_i|$. Thus, $T(\underline{X}) = (X_{(1)}, X_{(n)})$ is sufficient (or more minimally, $T = \max |X_i|$).

[2] Discrete Distribution Sufficiency:

Given $f_\theta(x) = \frac{c(\theta)}{x^2}$ for $x \in \{\theta + 1, \theta + 2, \dots\}$.

The joint probability mass function is:

$$\begin{aligned} f_\theta(\underline{x}) &= \prod_{i=1}^n \frac{c(\theta)}{x_i^2} \mathbb{I}(x_i \geq \theta + 1) \\ &= \frac{(c(\theta))^n}{\prod_{i=1}^n x_i^2} \prod_{i=1}^n \mathbb{I}(x_i \geq \theta + 1) \end{aligned}$$

The product of indicators is 1 if and only if all $x_i \geq \theta + 1$, which is equivalent to $\min(x_i) \geq \theta + 1$.

$$= \left[\frac{1}{\prod x_i^2} \right] \cdot [(c(\theta))^n \mathbb{I}(x_{(1)} \geq \theta + 1)]$$

By identifying $h(\underline{x}) = (\prod x_i^2)^{-1}$ and $g_\theta(t) = (c(\theta))^n \mathbb{I}(t \geq \theta + 1)$, we see that $T(\underline{X}) = X_{(1)}$ is a sufficient statistic for θ .

[3] Dependent Support Sufficiency:

$f(x) = \exp(i\theta - x) \mathbb{I}(x \geq i\theta)$.

The joint pdf is:

$$f_\theta(\underline{x}) = \prod_{i=1}^n e^{i\theta - x_i} \mathbb{I}(x_i \geq i\theta) = e^{\theta \sum i} e^{-\sum x_i} \prod_{i=1}^n \mathbb{I}\left(\frac{x_i}{i} \geq \theta\right)$$

The product of indicators is 1 if and only if $\min_i(\frac{x_i}{i}) \geq \theta$. Factorization:

$$f_\theta(\underline{x}) = \left[e^{-\sum x_i} \right] \cdot \left[e^{\frac{n(n+1)}{2}\theta} \mathbb{I}\left(\min_i \frac{x_i}{i} \geq \theta\right) \right]$$

Thus, $T(\underline{X}) = \min_{1 \leq i \leq n} \left(\frac{X_i}{i}\right)$ is sufficient for θ .

[4] Beta Distribution Sufficiency:

$f(x) \propto x^{\alpha-1}(1-x)^{\beta-1}$. Joint pdf:

$$f(\underline{x}) = \left(\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \right)^n \left(\prod x_i \right)^{\alpha-1} \left(\prod (1 - x_i) \right)^{\beta-1}$$

- (a) If β is known, the factor involving β is part of the constant or $h(\underline{x})$. The term depending on α involves $(\prod x_i)^\alpha$. Thus, $T = \prod_{i=1}^n X_i$ is sufficient for α .
- (b) If α is known, the term depending on β involves $(\prod (1 - x_i))^\beta$. Thus, $T = \prod_{i=1}^n (1 - X_i)$ is sufficient for β .
- (c) If both are unknown, the factorization separates terms for α and β . $T = (\prod_{i=1}^n X_i, \prod_{i=1}^n (1 - X_i))$ is jointly sufficient.

[5] Function of a Statistic:

Proof: Since T is sufficient, by the Factorization Theorem, $f_\theta(\underline{x}) = g_\theta(T(\underline{x}))h(\underline{x})$. Given $T = \psi(T^*)$, we can substitute:

$$f_\theta(\underline{x}) = g_\theta(\psi(T^*(\underline{x})))h(\underline{x})$$

Let $k_\theta(t) = g_\theta(\psi(t))$. Then $f_\theta(\underline{x}) = k_\theta(T^*(\underline{x}))h(\underline{x})$. Therefore, by the Factorization Theorem, T^* is also sufficient for θ .

[6] Minimal Sufficient Statistics:

To find the minimal sufficient statistic (MSS), we examine the ratio of likelihoods $\frac{L(\theta;\underline{x})}{L(\theta;\underline{y})}$. The statistic $T(\underline{X})$ is minimal sufficient if the ratio is independent of θ if and only if $T(\underline{x}) = T(\underline{y})$.

(a) $f_\alpha(x) = \frac{1}{\alpha} \exp\left(-\frac{x}{\alpha}\right), \quad x > 0.$

$$\frac{L(\alpha; \underline{x})}{L(\alpha; \underline{y})} = \frac{\alpha^{-n} \exp(-\frac{1}{\alpha} \sum x_i)}{\alpha^{-n} \exp(-\frac{1}{\alpha} \sum y_i)} = \exp\left(-\frac{1}{\alpha} \left(\sum x_i - \sum y_i\right)\right)$$

This ratio is independent of α iff $\sum x_i = \sum y_i$.

MSS: $T(\underline{X}) = \sum_{i=1}^n X_i$.

(b) $f_\beta(x) = \exp(-(x - \beta))\mathbb{I}(x > \beta).$

$$L(\beta; \underline{x}) = e^{-\sum x_i} e^{n\beta} \mathbb{I}(x_{(1)} > \beta)$$

$$\frac{L(\beta; \underline{x})}{L(\beta; \underline{y})} = e^{\sum y_i - \sum x_i} \frac{\mathbb{I}(\beta < x_{(1)})}{\mathbb{I}(\beta < y_{(1)})}$$

The ratio is independent of β iff the indicator functions match for all β , which implies $x_{(1)} = y_{(1)}$.

MSS: $T(\underline{X}) = X_{(1)}$.

(c) $f_{\alpha, \beta}(x) = \frac{1}{\alpha} \exp\left(-\frac{x-\beta}{\alpha}\right) \mathbb{I}(x > \beta).$

$$L(\alpha, \beta; \underline{x}) = \alpha^{-n} \exp\left(-\frac{\sum x_i - n\beta}{\alpha}\right) \mathbb{I}(x_{(1)} > \beta)$$

$$\frac{L(\underline{x})}{L(\underline{y})} = \exp\left(-\frac{1}{\alpha} \left(\sum x_i - \sum y_i\right)\right) \frac{\mathbb{I}(\beta < x_{(1)})}{\mathbb{I}(\beta < y_{(1)})}$$

For independence from α , we need $\sum x_i = \sum y_i$. For independence from β , we need $x_{(1)} = y_{(1)}$.

MSS: $T(\underline{X}) = (\sum_{i=1}^n X_i, X_{(1)})$.

(d) $f_{\mu, \sigma}(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right)$. Expanding the square: $(\log x - \mu)^2 = (\log x)^2 - 2\mu \log x + \mu^2$.

$$\frac{L(\underline{x})}{L(\underline{y})} \propto \exp\left(-\frac{1}{2\sigma^2} \left[\sum (\log x_i)^2 - \sum (\log y_i)^2\right] + \frac{\mu}{\sigma^2} \left[\sum \log x_i - \sum \log y_i\right]\right)$$

To be independent of μ and σ^2 , the coefficients must vanish: $\sum \log x_i = \sum \log y_i$ and $\sum (\log x_i)^2 = \sum (\log y_i)^2$.

MSS: $T(\underline{X}) = (\sum_{i=1}^n \log X_i, \sum_{i=1}^n (\log X_i)^2)$.

(e) $f_\theta(x) = \frac{1}{\theta} \mathbb{I}(|x| \leq \theta/2).$

$$L(\theta; \underline{x}) = \theta^{-n} \mathbb{I}(\max |x_i| \leq \theta/2) = \theta^{-n} \mathbb{I}(\theta \geq 2 \max |x_i|)$$

$$\frac{L(\theta; \underline{x})}{L(\theta; \underline{y})} = \frac{\mathbb{I}(\theta \geq 2 \max |x_i|)}{\mathbb{I}(\theta \geq 2 \max |y_i|)}$$

The ratio is constant (specifically 1) iff the cutoff points for the indicators are identical: $\max |x_i| = \max |y_i|$.

MSS: $T(\underline{X}) = \max_i |X_i|$.

(f) $f(x) \propto x^{\alpha-1} (1-x)^{\beta-1}$.

$$\frac{L(\underline{x})}{L(\underline{y})} = \left(\frac{\prod x_i}{\prod y_i}\right)^{\alpha-1} \left(\frac{\prod (1-x_i)}{\prod (1-y_i)}\right)^{\beta-1}$$

To be independent of α , $\prod x_i = \prod y_i$. To be independent of β , $\prod (1-x_i) = \prod (1-y_i)$.

MSS: $T(\underline{X}) = (\prod_{i=1}^n X_i, \prod_{i=1}^n (1-X_i))$.

[7] Unbiasedness of Squared Estimator:

Disprove. Let T be an unbiased estimator of θ , so $E[T] = \theta$. Consider $E[T^2]$. We know that $Var(T) = E[T^2] - (E[T])^2$.

$$E[T^2] = (E[T])^2 + Var(T) = \theta^2 + Var(T)$$

Since $Var(T) > 0$ for any non-degenerate random variable, $E[T^2] > \theta^2$. Thus, T^2 is a biased estimator of θ^2 .

[8] Normal Distribution $N(\theta, \theta)$:

(a) Show $T(\underline{X}) = \sum X_i^2$ is sufficient.

The pdf is $f(x) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{(x-\theta)^2}{2\theta}\right)$. Likelihood:

$$L(\theta) = (2\pi\theta)^{-n/2} \exp\left(-\frac{1}{2\theta} \sum_{i=1}^n (x_i - \theta)^2\right)$$

Expand the square: $\sum (x_i - \theta)^2 = \sum x_i^2 - 2\theta \sum x_i + n\theta^2$.

$$L(\theta) = (2\pi\theta)^{-n/2} \exp\left(-\frac{\sum x_i^2}{2\theta} + \sum x_i - \frac{n\theta}{2}\right)$$

Rearrange terms:

$$L(\theta) = \left[e^{\sum x_i}\right] \cdot \left[(2\pi\theta)^{-n/2} e^{-n\theta/2} e^{-\frac{\sum x_i^2}{2\theta}}\right]$$

We set $h(\underline{x}) = e^{\sum x_i}$ (depends only on data) and $g_\theta(t) = (2\pi\theta)^{-n/2} e^{-n\theta/2} e^{-t/2\theta}$. Since the likelihood factors this way, $T(\underline{X}) = \sum_{i=1}^n X_i^2$ is sufficient.

(b) Prove $U(\underline{X}) = 15T^2 + 8T - 23$ is sufficient.

Since $T(\underline{X})$ is sufficient, any one-to-one function of T is also sufficient. Consider the function $\psi(t) = 15t^2 + 8t - 23$. Since $X_i \sim N(\theta, \theta)$ with $\theta > 0$, $T = \sum X_i^2$ is a non-negative random variable ($T \geq 0$). For $t \geq 0$, the quadratic function $15t^2 + 8t - 23$ is strictly increasing (derivative $30t + 8 > 0$). Thus, the mapping is one-to-one on the support of T . Therefore, $U(\underline{X})$ is sufficient for θ .

[9] Uniform Unbiased Estimators:

For $U(0, \theta)$, the pdf of $X_{(n)}$ is $f_n(x) = \frac{nx^{n-1}}{\theta^n}$.

$$E[X_{(n)}] = \int_0^\theta x \cdot \frac{nx^{n-1}}{\theta^n} dx = \frac{n}{\theta^n} \left[\frac{x^{n+1}}{n+1} \right]_0^\theta = \frac{n}{n+1} \theta$$

Thus, $E\left[\frac{n+1}{n} X_{(n)}\right] = \theta$. Unbiased.

Also, $E[\bar{X}] = \mu = \theta/2$. Thus, $E[2\bar{X}] = 2(\theta/2) = \theta$. Unbiased.

[10] Normal $N(\theta, \theta^2)$ Unbiasedness:

Let $T_1 = \sum X_i$. $T_1 \sim N(n\theta, n\theta^2)$. $E[T_1^2] = Var(T_1) + (E[T_1])^2 = n\theta^2 + (n\theta)^2 = n\theta^2 + n^2\theta^2 = n(n+1)\theta^2$. Thus, $E\left[\frac{T_1^2}{n(n+1)}\right] = \theta^2$. Unbiased.

Let $T_2 = \sum X_i^2$. $E[X_i^2] = Var(X_i) + (E[X_i])^2 = \theta^2 + \theta^2 = 2\theta^2$. $E[T_2] = \sum E[X_i^2] = 2n\theta^2$. Thus, $E\left[\frac{T_2}{2n}\right] = \theta^2$. Unbiased.

[11] Poisson Unbiased Estimator for $\theta e^{-2\theta}$:

We need an estimator T such that $E[T] = \theta e^{-2\theta}$. Consider the sample points X_1 and X_2 . $P(X_1 = 1, X_2 = 0) = P(X_1 = 1)P(X_2 = 0) = \left(\frac{e^{-\theta}\theta^1}{1!}\right)\left(\frac{e^{-\theta}\theta^0}{0!}\right) = \theta e^{-\theta} \cdot e^{-\theta} = \theta e^{-2\theta}$. Define the estimator $T(\underline{X})$:

$$T(\underline{X}) = \begin{cases} 1 & \text{if } X_1 = 1, X_2 = 0 \\ 0 & \text{otherwise} \end{cases}$$

Then $E[T] = 1 \cdot P(X_1 = 1, X_2 = 0) = \theta e^{-2\theta}$.

[12] Bernoulli Limiting Unbiasedness:

$X_i \sim B(1, \theta)$, so $E[\sum X_i] = n\theta$.

(a) $E[T] = E\left[\frac{\frac{1}{2}\sqrt{n} + \sum X_i}{n + \sqrt{n}}\right] = \frac{\frac{1}{2}\sqrt{n} + n\theta}{n + \sqrt{n}}$. This is clearly not equal to θ for finite n .

(b) Limit as $n \rightarrow \infty$: Divide numerator and denominator by n :

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{2\sqrt{n}} + \theta}{1 + \frac{1}{\sqrt{n}}} = \frac{0 + \theta}{1 + 0} = \theta$$

Thus, it is unbiased in the limit.

[13] Location Family Unbiasedness:

Given $f(\theta - z) = f(\theta + z)$ (symmetry of pdf) and $T(\underline{x} + h) = T(\underline{x}) + h$ (translation equivariance) and $T(-\underline{x}) = -T(\underline{x})$ (odd function). Let $Y_i = X_i - \theta$. Then Y_i has a distribution symmetric about 0. $T(\underline{X}) = T(\underline{Y} + \theta) = T(\underline{Y}) + \theta$. $E[T(\underline{X})] = E[T(\underline{Y})] + \theta$. We need to show $E[T(\underline{Y})] = 0$. Since the joint pdf of $\underline{Y}$ is symmetric and T is an odd function ($T(-\underline{y}) = -T(\underline{y})$), the integral over the symmetric domain vanishes.

$$E[T(\underline{Y})] = \int T(\underline{y}) \prod f(y_i) d\underline{y} = 0$$

Therefore, $E[T(\underline{X})] = 0 + \theta = \theta$. Unbiased.